

PGT-Net: Prior Guided Transformer for Hyperspectral Image Denoising

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Abstract—Transformer-based denoising methods have achieved promising performance in recent works. However, most of them suffer from two major deficiencies. First, model interpretability is constrained by the widely accepted black-box principle. Second, denoising performance is limited by the inability of current Transformer-based techniques to model both local details and global context simultaneously. To address these issues, we propose an innovative Prior-Guided Transformer (PGT) for image denoising, which combines the advantages of hyperspectral image (HSI) priors and advanced network architectures. Specifically, we design two sub-networks: a physical prior-guided sub-network (P-Net) and an implicit prior-guided sub-network (I-Net). For P-Net, an Adaptive-Frequency Filtering Unit (AFU) is added to model the low-rank structure and suppress noisy components based on frequency distribution. For I-Net, we develop a Scaling Attention Unit (SAU) and a Multi-Modal Fusion Unit (M-MFU) to efficiently encode non-local similarity between tokens and to fuse spatial-spectral features from different layers for enhanced texture restoration. Extensive experiments demonstrate that PGTNet achieves state-of-the-art performance on both synthetic and real-world image denoising tasks.

Index Terms—Hyperspectral image denoising, prior-guided, transformer.

I. INTRODUCTION

HYPERSPECTRAL images, with their high spectral resolution and a vast number of bands, are frequently used in remote sensing [1], [2], medicine [3], agriculture [4], and other fields [5], [6]. However, hyperspectral images inevitably suffer from various corruptions and degenerations, which can

impede further data processing and interpretation. Therefore, how to effectively restore the tiny textures and details from the noise is an important issue in hyperspectral image processing.

A. Related Work

This section briefly introduces recent works on hyperspectral denoising as well as prior-guided deep learning. Hyperspectral denoising has always been a challenging problem. Therefore, effectively incorporating priors from clean images is key to improve denoising performance. In general, existing methods can be roughly divided into two categories: model-based approaches and learning-based approaches.

1) *Model-based approaches.*: In recent years, scholars have developed various effective regularization techniques for denoising, including spectral low-rankness [7], [8], spatial redundancy [9], [10] and nonlocal similarity [11], [12]. Among these approaches, matrix/tensor-based low-rank methods, particularly LRMR [13] and LRTV [14], have achieved state-of-the-art performance owing to their thorough modeling of intrinsic hyperspectral imaging (HSI) properties.

2) *Learning-based approaches.*: Driven by advances in deep learning, researchers are focusing on building denoising networks that can learn implicit priors to reconstruct fine-grained textures from noisy HSIs. For example, SST [15] reshapes the height and width dimensions into one, then calculates attention across all spectral vectors. TRQ3DNet [16] incorporates a shifted windowing scheme to model non-local spatial relations. HSDT [17] builds upon the concept of attentive information flow by modifying the attention blocks to directly examine the full 3D spectral-spatial space. Diff-Unmix [18] proposes a self-supervised diffusion denoising model that achieves HSI denoising through spectral unmixing and abundance generation. However, most of these approaches heavily rely on large-scale training datasets, resulting in low generalization and a significant amount of parameter redundancy.

3) *Prior-based approaches.*: In recent research, many scholars increasingly adopt prior-based approaches that combine model-driven insights with data-driven learning to improve both interpretability and performance. Among them, deep unfolding networks (DUNs) offer a structured way to embed priors into deep models, and have been applied to many image tasks. For the discussed denoising task, T3SC [19] and SMDS [20] are two representative DUNs. However, their limited capacity to capture global spectral correlations restricts their effectiveness in leveraging both HSI structural

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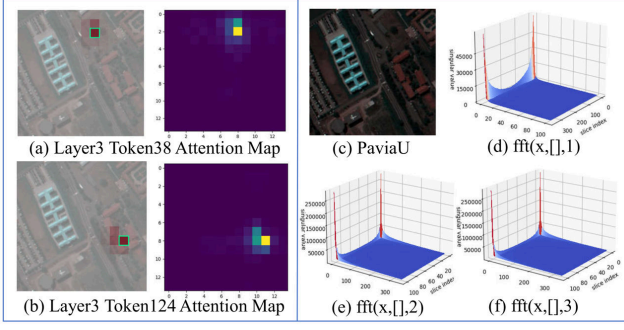


Fig. 1. (a)-(b) Implicit Priors: Attention maps from the 3rd ViT layer show that only nearby tokens (green box) contribute significantly. (c)-(f) Physical Priors: FFT decomposition along spatial and spectral dimensions of the HSI reveals pronounced low-rank structures exist in all three directions.

priors and learned features. Thus, a key challenge remains in designing an effective denoiser that can efficiently fuse the physical structure of HSIs with learned data priors for improved denoising performance. The main contributions of this work can be summarized as follows:

- An end-to-end denoising PGT model that combines the advantages of physical prior information and advanced network architecture for interpretable and robust HSI denoising.
- Two novel sub-networks are proposed based on a thorough priors analysis. P-Net uses frequency decomposition based on physical priors to suppress noise, while I-Net models implicit priors to preserve global-local texture features with low computation.
- Experimental results on both synthetic and real HSI datasets confirm that the proposed model achieves state-of-the-art performance, particularly in terms of texture detail preservation and model convergence.

II. PGT HSIs DENOISING MODEL

In this section, we introduce our prior-guided denoising model and also its associated optimisation procedure.

A. Model Formulation

Briefly, the imaging degradation model of the hyperspectral remote sensing system can be formulated as

$$\mathcal{Y} = \mathcal{X} + \mathcal{E} \quad (1)$$

where $\mathcal{Y}$ denotes a noisy observed HSI containing M pixels and N bands, $\mathcal{X}$ indicates the ideal noise-free data, and $\mathcal{E}$ represents the noise item. This is an ill-posed inverse problem, and the prior knowledge on $\mathcal{X}$ is very important to solve the problem. On the one hand, for low rank physical priors, we adopt dictionary learning approach, combined with a tensor nuclear norm regularization $\|\mathcal{X}\|_{PP}$. On the other hand, for complex implicit priors, we introduce an data prior regularization $\|\mathcal{C}\|_{IP}$, which means the local-nonlocal feature correlations inherently modeled by deep neural networks. Therefore, the problem can be expressed as

$$\mathcal{Y} = \mathcal{D} * \mathcal{C} + \mathcal{E} \quad (2)$$

$$\min_{\mathcal{D}, \mathcal{C}, \mathcal{X}} \|\mathcal{C}\|_{IP} + \gamma \|\mathcal{X}\|_{PP} \quad (3)$$

$$\text{s.t. } \mathcal{D}^T * \mathcal{D} = \mathcal{I}, \|\mathcal{Y} - \mathcal{X}\|_F^2 < \varepsilon, \|\mathcal{X} - \mathcal{D} * \mathcal{C}\|_F^2 < \varepsilon$$

where $\mathcal{D} \in R^{N \times n}$ denotes the subspace basis tensor with $n \ll N$ to capture the spatial-spectral redundancy and $\mathcal{C} \in R^{n \times M}$ denotes the coefficient tensor. $*$ denotes the tensor t-product. $\|\mathcal{X}\|_{PP}$ denotes physical priors, which means the tensor nuclear norm enforced for the reconstructed clean tensor $\mathcal{X}$ to keep low-tubal-rank and $\mathcal{D}^T * \mathcal{D} = \mathcal{I}$ is the tensor orthogonal constraint that force the tensor basis of subspace to span the original high-dimensional space as much as possible.

B. Motivations and Contributions

The main challenge in denoising lies in balancing noise removal and the preservation of clean signals. To address this, we conduct an experimental analysis on the PaviaU dataset and propose PGT-Net.

1) *Implicit Priors.*: With the advancement of neural networks, increasingly powerful models are capable of learning highly complex and intricate mappings. Among them, Vision Transformer (ViT) [21] can capture global dependencies, achieving outstanding performance in vision tasks. However, as shown in Fig. 1 (a-b), ViT indeed learns detailed representations in the early layers, but with redundant spatial and temporal attention. To tackle this issue, we propose a scaling attention mechanism that dynamically adjusts the scope of attention computation, thereby capturing both local and non-local spatial similarities.

2) *Physical Priors.*: HSIs exhibit a highly distinctive structure, characterized by typical low-rank properties. Taking the HSI PaviaU, shown in Fig. 1 (c-f), as an example, we observe that slices along all modes are approximately low-rank, indicating clear spatial and spectral correlations both quantitatively and visually. Such inherent properties of HSIs can be effectively captured through explicit physical modeling via nuclear norm minimization [22]. Given that, we first introduce a tensor dictionary learning decomposition to reduce computational complexity. Then, we propose a subnetwork that leverages low-rank properties across all modes, effectively maintaining the low-rank structure and preserving the 3D spatial-spectral correlations of the data.

Following the above analysis, this paper proposes a Priors-guided Deep Unfolding Network (PGTNet) for interpretable denoising. Specifically, we first formulate a unified optimization model based on the degraded observation characteristics of HSIs and design a priors-guided algorithm to alternately solve the physical prior and implicit prior subproblems.

C. Model Optimization

The objective function in Eq. (3) has two unknown variables and each separate term of the proposed model can be effectively convex approximated. Therefore, the inclusion of the distributed alternating direction method of multipliers (ADMM) [23] algorithm produces a fast solution for the proposed model. The augmented lagrangian function $L(\bullet)$ is

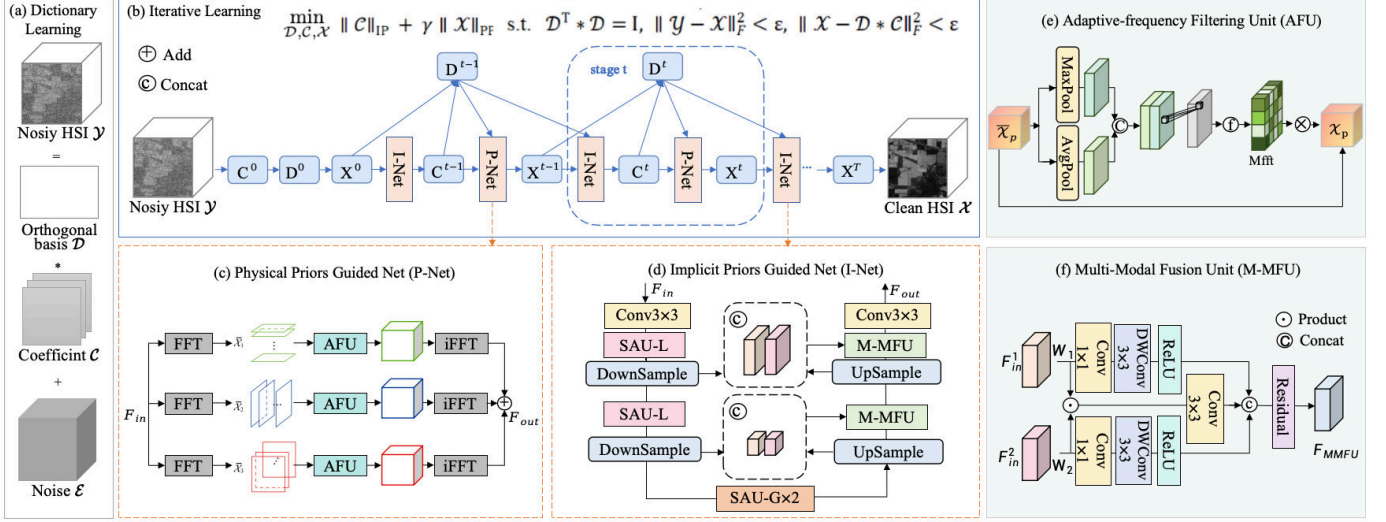


Fig. 2. Illustration of the PGTNet. (a) Dictionary learning details. (b) Overall architecture of PGTNet with T stages. Each stage contains a P-Net and an I-Net, corresponding to the Eq. (8) and the Eq. (9). (c) P-Net pipeline. (d) I-Net pipeline. (e) AFU details. (f) M-MFU details.

formulated as

$$L(\mathcal{C}, \mathcal{D}, \mathcal{X}, \mathcal{G}_1, \mathcal{G}_2) = \|\mathcal{C}\|_{IP} + \gamma \|\mathcal{X}\|_{PP} + \zeta_{\{I\}}(\mathcal{D}^T * \mathcal{D}) + \langle \mathcal{G}_1, \mathcal{Y} - \mathcal{X} \rangle + \langle \mathcal{G}_2, \mathcal{X} - \mathcal{D} * \mathcal{C} \rangle + \frac{\mu}{2} \|\mathcal{Y} - \mathcal{X}\|_F^2 + \frac{\mu}{2} \|\mathcal{X} - \mathcal{D} * \mathcal{C}\|_F^2 \quad (4)$$

where $\mathcal{G}_1$ and $\mathcal{G}_2$ are two augmented Lagrange multipliers, $\mu > 0$ is the ADMM penalty parameter, and $\zeta_{\{I\}}(\mathcal{D}^T * \mathcal{D})$ is an indicator function that is a transformation of the orthogonal constraint of the tensor basis. Next, we can obtain $\mathcal{D}$, $\mathcal{C}$ and $\mathcal{X}$ by solving the augmented lagrangian sub-function of Eq. (4) separately as following.

$$\mathcal{C}^{(t+1)} = \arg \min_{\mathcal{C}} \|\mathcal{C}\|_{IP} + \frac{\mu}{2} \left\| \mathcal{D}^{(t)} * \mathcal{C} - \mathcal{X}^{(t)} - \frac{\mathcal{G}_2^{(t)}}{\mu} \right\|_F^2 \quad (5)$$

$$\mathcal{D}^{(t+1)} = \arg \min_{\mathcal{D}, \mathcal{D}^T * \mathcal{D}} \frac{\mu}{2} \left\| \mathcal{D} * \mathcal{C}^{(t+1)} - \mathcal{X}^{(t)} - \frac{\mathcal{G}_2^{(t)}}{\mu} \right\|_F^2 \quad (6)$$

$$\mathcal{X}^{(t+1)} = \arg \min_{\mathcal{X}} \gamma \|\mathcal{X}\|_{PP} + \frac{\mu}{2} \left\| \mathcal{X} - \mathcal{Y} - \frac{\mathcal{G}_1^{(t)}}{\mu} \right\|_F^2 + \frac{\mu}{2} \left\| \mathcal{D}^{(t+1)} * \mathcal{C}^{(t+1)} - \mathcal{X} - \frac{\mathcal{G}_2^{(t)}}{\mu} \right\|_F^2 \quad (7)$$

The solution of the optimization problem of $\mathcal{C}$ in Eq. (5) and $\mathcal{X}$ in Eq. (5) can be corresponded to a denoising problem [24], [25], [26], [27]. $\|\mathcal{C}\|_{IP}$ is the implicit data prior term used to constrain the solution space. $\|\mathcal{X}\|_{PP}$ is the tensor nuclear norm to constraint the solution space, and λ is a trade-off parameter. In this work, we elaborate I-Net to approximate $\mathcal{C}^{(t+1)}$, which prominently mines global and local similarity for more robust denoising. We apply P-Net to approximate the proximal operator $\mathcal{X}^{(t+1)}$, which preserving the intrinsic low-rank structure and 3D spatial-spectral relationships.

$$\mathcal{C}^{(t+1)} = f_{I\text{-Net}}(\hat{\mathcal{C}}^{(t+1)}) \quad (8)$$

$$\mathcal{X}^{(t+1)} = f_{P\text{-Net}}(\hat{\mathcal{X}}^{(t+1)}) \quad (9)$$

The optimization problem of $\mathcal{D}$, we apply the T-SVD technique [28] to obtain the following closed-form solution:

$$\{\mathcal{U}, \mathcal{V}\} = \text{T-SVD} \left(\left(\mathcal{X}^{(t)} + \frac{\mathcal{G}_2^{(t)}}{\mu} \right) * (\mathcal{C}^{(t+1)})^T \right) \quad (10)$$

$$\mathcal{D}^{(t+1)} = \mathcal{U} * \mathcal{V}^T$$

where $\mathcal{U}$ and $\mathcal{V}$ are the left and right singular tensor.

The update of augmented Lagrange multipliers $\{\mathcal{G}_1, \mathcal{G}_2\}$:

$$\mathcal{G}_1^{(t+1)} = \mathcal{G}_1^{(t)} + \mu (\mathcal{Y} - \mathcal{X}^{(t+1)})$$

$$\mathcal{G}_2^{(t+1)} = \mathcal{G}_2^{(t)} + \mu (\mathcal{X}^{(t+1)} - \mathcal{D}^{(t+1)} * \mathcal{C}^{(t+1)}) \quad (11)$$

Overall, the denoising procedure can be summarized in Algorithm 1. Thanks to end-to-end training, the algorithm can be alternatively regarded as a forward process of a DNN and the construction error between estimated HSIs and ground-truth HSIs can be back propagated to guide the network learning. To this end, we construct a deep neural network based on the denoising procedure in Algorithm 1 as will be presented in the next subsection.

Algorithm 1 PGT for HSIs denoising procedure
Input: Noisy HSI $\mathcal{Y}$.
Output: Denoised HSI $\mathcal{X}$.
While not converged do
Compute $\mathcal{C}^{(t+1)}$ with Eq. (8).
Compute $\mathcal{X}^{(t+1)}$ with Eq. (9).
Compute $\mathcal{D}^{(t+1)}$ with Eq. (10).
Compute $\{\mathcal{G}_1^{(t+1)}, \mathcal{G}_2^{(t+1)}\}$ with Eq. (11).
End while
Output the estimated clean hyperspectral image $\hat{\mathcal{X}}$.

TABLE I
PGT MODEL

D. PGTNet

As illustrated in Fig. 2, we unfold all the steps of PGT in Algorithm 1 as network layers to solve the denoising problem. PGTNet consists of multiple stages.

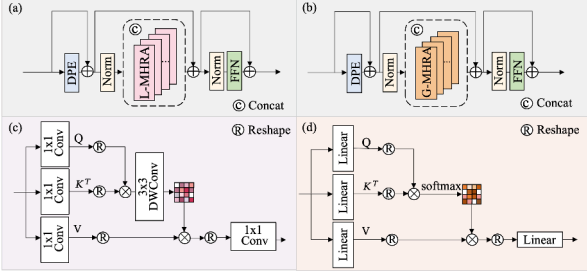


Fig. 3. Architecture of Scaling Attention Unit (SAU). (a)-(b) SAU-L/G details. (c)-(d) Local/Global Multi-Head Relation Aggregator pipeline (L/G-MHRA).

1) *Physical Priors Guided Sub-Network (P-Net)*: Fourier transform offers a natural way to decompose feature maps into distinct frequency components [29]. Motivated by this, we design a P-Net to approximate the closed-form solution of the data subproblem at the gradient descent step as shown in Eq. (9). As shown in Fig. 2 (c), in the k -th stage, the data module takes $\hat{\mathcal{X}}^{(t+1)}$ as its input and produces $\mathcal{X}^{(t+1)}$ as its output. First, a 2D Fast Fourier Transform (FFT) projects spatial features into the frequency domain. Second, an Adaptive Frequency Filtering Unit (AFU) selectively performs adaptive weighting across different frequency components, emphasizing texture-related frequencies while suppressing irrelevant noise. Finally, a 2D inverse FFT (iFFT) reconstructs the enhanced features back into the original latent space, preserving essential structures with improved clarity.

Adaptive-frequency filtering Unit (AFU): As shown in Fig. 2(e), given an input modeled as a third-order tensor, the AFU module first applies directional FFT along the spatial dimensions to capture frequency-domain priors that reflect intrinsic structural correlations. After FFT, the frequency information of $\mathcal{X}$ is aggregated over channels using both average-pooling and max-pooling operations along the channel axis, generating two frequency descriptors, denoted by $F_{\text{avg}}^{\text{fft}} \in R^{M \times N \times 1}$ and $F_{\text{max}}^{\text{fft}} \in R^{M \times N \times 1}$. These descriptors summarize the global frequency responses at each spatial location, respectively while the spatial discrimination ability is still preserved. The concatenated descriptors are then fed into a large-kernel 7×7 convolution layer to generate a spatial mask, enabling the model to selectively enhance informative frequency regions while attenuating noise-dominated areas.

2) *Implicit Priors Guided Sub-Network (I-Net)*: In HSI denoising, capturing both local and global spatial dependencies is crucial due to the complex data structure. Motivated by this, we design a I-Net to approximate the closed-form solution of the data subproblem at the gradient descent step Eq. (8). Given the inputs $\hat{\mathcal{C}}^{(t+1)}$ as the module input and outputs $\mathcal{C}^{(t+1)}$ as the module output. As depicted in Fig. 2(d), I-Net adopts a Uniformer-style architecture with skip connections to jointly model local and global features. It integrates two key components: a Scaling Attention Unit (SAU) in the encoder and a Multi-Modal Fusion Unit (M-MFU) in the decoder. SAU enhances self-attention by introducing local and global tokens to capture short- and long-range dependencies, while M-MFU

fuses multi-scale features for better texture detail restoration. Down- and up-sampling are implemented using strided and transposed convolutions, respectively. Subsequently, the details will be explained.

Scaling Attention Unit (SAU): As shown in Fig. 3, the encoding unit consists of three key modules: Dynamic Position Embedding (DPE), Multi-Head Relation Aggregator (MHRA), and Feed-Forward Network (FFN). Given an input feature map $F_{\text{in}} \in R^{h \times w \times n}$, where h , w , and n represent the height, width, and number of channels, respectively. Firstly, SAU uses depth-wise convolution with zero padding (DPE) to encode visual information and inject dynamic positional cues into all tokens. The core of SAU lies in the Multi-Head Relation Aggregator (MHRA), which adaptively captures contextual dependencies via salcing self-attention by flexibly designing its contextual token affinity in the shallow and deep layers. As Eq. (12), where $R_n(\cdot)$ refer to the relation aggregator in the n -th head. $ATT_n(\cdot)$ is an affinity matrix that describes token relation, while $U \in R^{n \times n}$ is a learnable fusion matrix.

$$\begin{aligned} \text{MHRA}(F_{PE}) &= \text{Concat}(R_1(F_{PE}); R_2(F_{PE}); \dots; R_N(F_{PE}))U \\ R_n(F_{PE}) &= \text{ATT}_n V_n(F_{PE}) \end{aligned} \quad (12)$$

For shallow layers, SAU-L performs local self-attention by dividing the feature map into the small neighborhood as Eq. (13) and Fig. 3(c). This mechanism, similar to the design of a convolution filter, enables efficient learning of short-range dependencies and reduces computational complexity.

$$\text{ATT}_n^{\text{local}} = \text{DWConv}(Q_n K_n) \quad (13)$$

For deep layers, SAU-G adopts a global contextual modeling strategy by computing token affinity based on content similarity among all tokens, as shown in Eq. (14) and Fig. 3(d). This formulation allows each token to aggregate information from the entire spatial context, thereby enhancing semantic-level understanding.

$$\text{ATT}_n^{\text{global}} = (Q_n K_n) \quad (14)$$

Finally, a Feed-Forward Network (FFN) consisting of two linear layers with a GeLU activation is applied to enhance feature diversity and increase representational capacity.

Multi-Modal Fusion Unit (M-MFU): As shown in Fig. 2(f), after the encoding stage, which captures multi-level features, the Multi-Modal Fusion Unit (M-MFU) incorporates both shallow and deep features. Firstly, M-MFU encodes channel-wise context by applying 1×1 convolutions followed by 3×3 depth-wise convolutions. The Hadamard product then models the fine-grained interaction. Finally, the interaction features and attended features are concatenated and passed through a Residual block. The feature effectively captures both the global and local context for the current spatial resolution.

III. EXPERIMENT RESULTS

In this section, we introduce the setting of training data and testing data. What's more, both simulated-data experiments and real-data experiments are involved to verify the effectiveness of the pro-posed method.

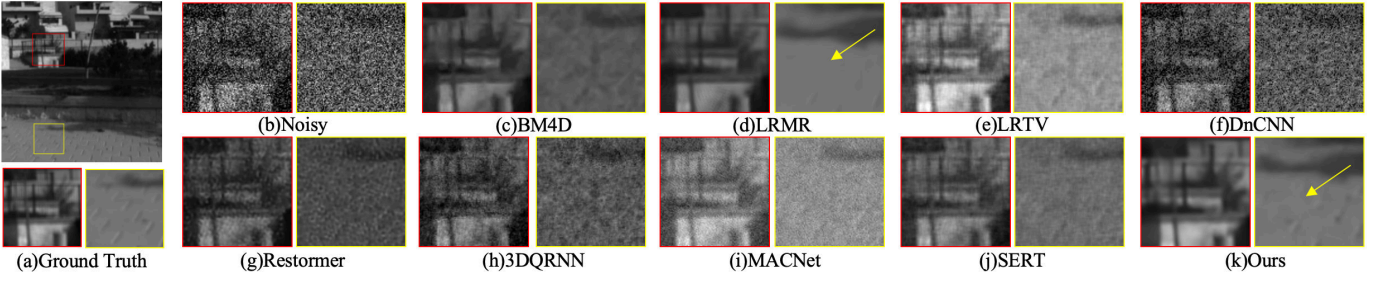


Fig. 4. Simulated Gaussian noise removal results on the ICVL dataset ($\sigma = 50$). (a) Ground truth image; (b) error map of the noisy image; (c) result after Gaussian noise addition; (d–l) denoising results from different models.

Case	Index	Noisy	Model-based			CNN-based			Transformer-based		
			BM4D	LRMR	LRTV	DnCNN	3DQRN	T3SC	SERT	Restorme	Ours
Case1	PSNR/SSIM	17.71/0.13	37.39/0.89	29.05/0.53	35.23/0.92	29.23/0.79	38.87/0.90	38.99/0.90	40.07/0.96	33.45/0.88	42.66/0.99
Case2	PSNR/SSIM	18.07/0.14	36.82/0.88	29.05/0.53	29.77/0.80	29.77/0.80	37.56/0.90	37.43/0.90	38.94/0.95	31.45/0.90	41.39/0.99
Case3	PSNR/SSIM	18.32/0.15	35.38/0.87	28.37/0.52	29.02/0.76	29.02/0.76	36.78/0.90	36.94/0.90	38.76/0.95	32.59/0.86	41.54/0.99
Case4	PSNR/SSIM	14.41/0.08	26.28/0.50	24.85/0.43	21.79/0.48	21.79/0.48	37.65/0.90	37.54/0.90	39.26/0.90	29.41/0.62	40.90/0.92
Case5	PSNR/SSIM	14.03/0.08	22.68/0.39	24.10/0.42	20.99/0.36	20.99/0.35	36.10/0.92	35.79/0.92	38.65/0.94	28.43/0.65	39.90/0.98

TABLE II
QUANTITATIVE RESULTS OF DIFFERENT METHODS IN COMPLEX NOISE CASE ON ICVL DATASET

A. Training Experimental Settings

1) *Training Data Set.*: We conduct several experiments using data from ICVL hyperspectral dataset [30], which comprise 201 images at a resolution of 1392×1300 over 31 spectral bands. We use 100 images for training, 1 image for validation, and 40 images for testing. The training set is expanded by cropping the photos into $64 \times 64 \times 31$ patch pairs. Each image is standardized into $[0, 1]$ before conducting experiments.

2) *Noise setting.*: We consider five different noise settings for training the model.

Case 1: Non-i.i.d. Gaussian Noise with randomly chosen from 30 to 70.

Case 2: Gaussian + Stripe Noise. Gaussian Noise in case 1, ten bands are randomly selected to add stripe noise, and the number of stripes in each band is 5% to 15% of columns.

Case 3: Gaussian + Deadline Noise. Gaussian Noise in case 1, ten bands are chosen randomly to add deadline noise. The number of deadlines in each band is 5% to 15% of columns.

Case 4: Gaussian + Impulse Noise. Gaussian Noise in case 1, ten bands are randomly selected to add impulse noise with different intensities, and the percentage of impulse ranges from 10% to 70%.

Case 5: Mixture Noise. Gaussian Noise with at least one kind of the other four noise mentioned above.

3) *Competing Methods.*: For traditional approaches, we compare our model with filtering-based method, such as BM4D [31], and low-rank-based methods, including LRMR [13] and LRTV [14]. For deep learning approaches, we compare our model with CNN-based methods, including DnCNN [32] and 3DQRNN [33], as well as Transformer-based methods: Restormer [34], SERT [35], TRQ3DNet [16], SST [15], HSDT [17] and Diff-unmix [18]. We also compared our model with deep unfolding-based methods T3SC [36] and SMDS[37]. DnCNN and Restormer are performed in a band-

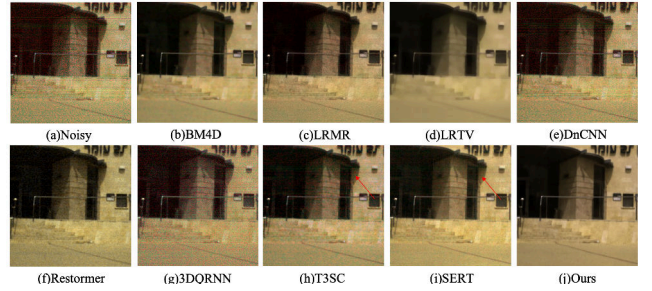


Fig. 5. Denoising results on the ICVL dataset with simulated complex noise (Case5: Mixture Noise) The false-color images are generated by combining bands 23, 18, and 8.

by-band manner. We tried our best to manually tune the parameters of model-based methods to the optimal one.

4) *Evaluation Indexes.*: In order to evaluate the denoising performance of simulated experiments in both the spatial domain and spectral domain, peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) are introduced as quantitative criteria. Larger PSNR and SSIM imply better denoising.

5) *Implementation Detail.*: Hyper-parameter values were empirically set to make network learning fast yet stable. The batch size is set to 16. We used the Adam algorithm as the optimizer. All experiments were performed on a PC with an Intel(R) Xeon(R) Gold 5218R CPU, and an NVIDIA 2080Ti GPU.

B. Simulated-Data Assessments

1) *Testing Data Set.*: We design two different scenarios: gaussian denoising and complex denoising to evaluate the denoising performance of PGTNet.

2) *Verification on Gaussian Denoising.*: In the Gaussian Noise Case, from Fig. 4(b), it can be observed that the image is contaminated by speckled Gaussian noise. Compared

with the ground truth in Fig. 4(a), although LRMR achieves cleaner denoising, it fails to preserve the texture of the floor tiles effectively. In contrast, our method can not only remove the gaussian noise well, but also preserve the structure and details better, so as to obtain better visual reconstruction results. The CNN-based algorithm struggles to capture long-range dependency, due to the limited receptive field, while our PGNet can achieve the best or the second best index in most bands thanks to the I-Net to consider the spatial-spectral correlation of noisy HSI.

TABLE III
DENOISING RESULTS AND COMPLEXITY OF DIFFERENT
MODULE SETTINGS

Dictionary Learning	Physical Prior	Implicit Prior	PSNR	SSIM	Params (M)	Flops (G)
✓	✓	✓	43.144	0.991	1.49	23.05
✓	×	✓	37.412	0.798	1.32	22.82
✓	✓	×	41.377	0.883	1.15	15.79
×	✓	✓	42.349	0.969	1.59	27.41

3) *Verification on Complex Denoising.*: In the complex noise case, denoising quantization results are shown in Table II. From Table II, our method achieves superior performance, outperforming others by more than 1 dB in PSNR. Compared with methods based on Transformer (Restormer and SERT), our method can restore clean texture information and suppress noise thanks to the AFU. The synthetic color image is given in Fig. 5, using red color to mark some apparent degradations. It can be easily observed that although the competing methods LRTV, LRMR, and SERT can obtain cleaner denoising results, but still contains some noise. In contrast, our method can produce cleaner and sharper results.

C. Real-Data Assessments

1) *Testing Data Set.*: To evaluate the effectiveness of PGNet in real-world denoising scenarios, we conducted experiments on two real-world datasets: Urban [38] and Indian Pines [39]. The Urban dataset, captured by the Hyperion sensor, has dimensions of $307 \times 307 \times 210$ and is primarily contaminated by a combination of stripe noise, dead lines, and Gaussian noise. The Indian Pines dataset, acquired by the AVIRIS sensor with dimensions of $145 \times 145 \times 224$, suffers from severe Gaussian noise, salt-and-pepper noise, and dead lines in several bands due to environmental interference.

2) *Verification on Real Sensed Denoising.*: Fig. 6 and Fig. 7 show that for most model-based or CNN-based methods, only slight suppression of external color noise is observed, while many noise artifacts still remain. Deep unfolding-based and Transformer-based methods show improved results but still exhibit residual noise or blurring, as highlighted by the yellow arrows in the figures. Specifically, although T3SC is designed based on a Gaussian noise assumption, it still shows noticeable residual noise in real-world scenarios, particularly evident in Fig. 7. Methods such as HSDT, SST and TRQ3DNet primarily rely on spatial attention mechanisms while overlooking spectral correlations, leading to suboptimal denoising performance. Diff-Unmix introduces the diffusion

framework into HSI denoising tasks, but it may also result in the generation of pseudo-noise. In contrast, PGNet effectively combines spatial-spectral priors with a robust network architecture, achieving superior denoising performance while preserving the original image features.

IV. DISCUSSION

A. Impact of Key Components

In this section, we evaluate denoising performance and computational complexity of key components in our model design under a blind Gaussian denoising setting. As shown in Table III, the physical prior brings the most significant improvement, as it effectively suppresses noise and preserves structural details. Dictionary learning also improves performance with minimal computational cost by enhancing the model adaptability to data. The implicit prior helps capture long-range dependencies, improving accuracy, but introduces considerable computational overhead due to attention mechanisms. Notably, combining priors and dictionary learning achieves a good balance between performance and efficiency.

B. Effectiveness of Physical Priors Guided Network (P-Net) Key Components

TNN[44] is a well-known and widely used traditional method for modeling low-rankness. Therefore, we compare TNN with our proposed AFU module in the P-Net framework. Due to TNN only examines low rank in the spectral dimension, stripe noise will be retained. Unlike TNN, AFU captures low-rank structures across all three dimensions and adaptively computes correlations without requiring manually set parameters. As shown in Table V, AFU achieves a PSNR of 43.144 and SSIM of 0.991 with a runtime of just 0.43s, outperforming TNN in both accuracy and efficiency.

C. Effectiveness of Implicit Priors Guided Network (I-Net) Key Components

FFN: It shows that FFN in our model can further mix token context at each position to boost classification accuracy. Removing FFN significantly degrades performance, where PSNR and SSIM drop from 43.144 to 40.917 and from 0.991 to 0.934, respectively.

DPE: With dynamic position embedding, our PGNet obviously improves 2.147 on PSNR but 0.094 on SSIM. It shows that via encoding the position information, our DPE can maintain spatial and temporal order, thus contributing to better representation learning.

MMFU: Similarly, PSNR and SSIM improve from 41.917 to 43.144 and from 0.934 to 0.991, respectively by adding the M-MFU. This is because the M-MFU can efficiently fuse the multi-level features from both low-level spatial features and high-level semantic context.

MHRA: More importantly, we investigate the configuration of local and global SAU block stage by stage, in order to verify the effectiveness of our network. As shown in row1 in Table VI, when we only use local MHRA (LLLL), the computation cost is light. However, the accuracy is largely dropped (2.052

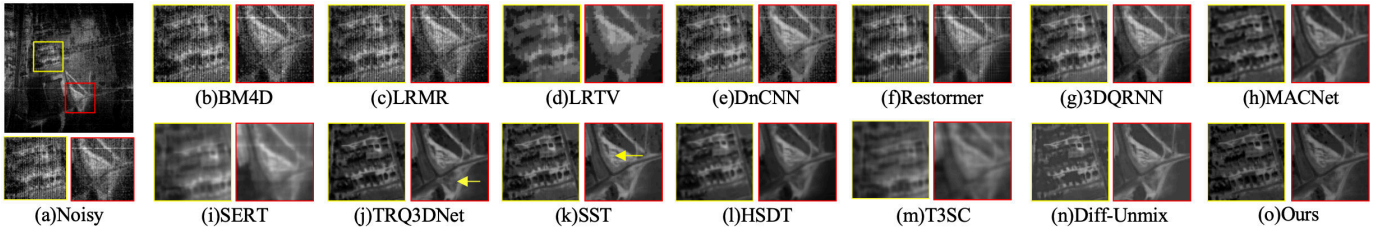


Fig. 6. Real noise removal results on Urban dataset scene 1

Metric	DnCNN	Restormer	3DQNN	SST	HSDT	TRQ3DNet	SERT	T3SC	SMDs	MACNet	Diff-Unmix	PGTNet
Parameters (M)	0.55	26.11	0.86	4.14	2.09	0.78	1.91	0.83	4.98	0.43	8.56	1.49
PSNR (dB)	20.99	28.43	36.10	34.89	38.93	38.67	38.65	35.79	38.24	38.20	33.06	40.19
Runtime (s)	2.85	0.60	1.41	1.80	0.97	0.85	0.72	0.95	8.32	3.44	24.17	0.43

TABLE IV
THE PARAMETERS, PSNR AND RUNTIME OF DIFFERENT DL-BASED DENOISING METHODS

TABLE V
DENOISING RESULTS AND COMPLEXITY OF P-NET KEY COMPONENTS

Setting	PSNR	SSIM	Time(s)
with AFU	43.144	0.991	0.43
with TNN	38.721	0.863	20.78

and 0.105 on PSNR and SSIM) due to the lack of capacity for learning long-term dependency. When we gradually replace local MHRA with global MHRA, the accuracy becomes better as expected. Unfortunately, the accuracy dramatically drops with a heavy computation load when all the layers apply global MHRA (GGGG), i.e., 0.112 on SSIM. It is mainly because that, without local MHRA, the network lacks the ability to extract detailed representations, leading to severe model overfitting with redundant attention for HSIs data. Empirically, the LLGG configuration achieves the best trade-off between denoising performance and computational efficiency.

TABLE VI
DENOISING RESULTS AND COMPLEXITY OF I-NET KEY COMPONENTS

FFN	DPE	MHRA (Type)	MMFU	PSNR	SSIM	Params (M)	Flops (G)
✓	✓	LLGG	✓	43.144	0.991	1.49	23.1
×	✓	LLGG	✓	41.034	0.921	1.42	22.1
✓	×	LLGG	✓	40.997	0.897	1.49	22.6
✓	✓	LLGG	×	41.917	0.934	1.37	20
✓	✓	LGGG	✓	41.541	0.932	1.52	21.6
✓	✓	LLLG	✓	41.387	0.897	1.57	17.4
✓	✓	LLLL	✓	41.092	0.886	1.61	17.4
✓	✓	GGGG	✓	42.037	0.879	1.29	41.2

D. Assessment of Model Size

This section compares the model size for different algorithms. As shown in Table IV, thanks to the scaling attention mechanism (SAU), our method significantly reduces the

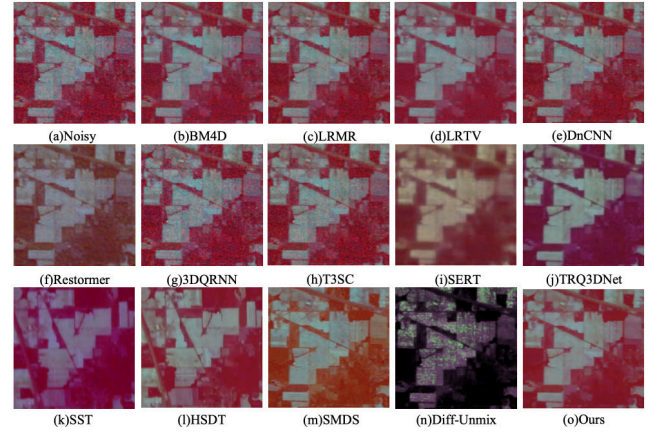


Fig. 7. False-color images of denoising results on the Indian Pines dataset with Real noise. (R:5, G:20, B:30)

number of parameters required compared to other transformer-based methods. Despite having smaller models, CNN-based methods struggle to capture long-range dependencies and restore texture features, leading to poor performance in complex noise scenarios. In summary, the proposed method offers efficient computation while ensuring competitive performance. For inference time cost, we compared the different methods in mixture complex noise removal task. Among DL-based methods, PGTNet spends the least time, benefiting from the dictionary learning mechanism that reduces the feature dimensions for attention operations. In summary, our PGTNet achieves state-of-the-art performance with high computational efficiency.

V. CONCLUSION

Although various denoising methods exist, most fail to fully exploit the priors of hyperspectral noisy images. We propose PGTNet, a denoising network that integrates model priors with advanced network design. It consists of two subnetworks: P-Net, guided by physical priors and equipped with an adaptive-frequency filtering unit (AFU) to model low-rank spatial-spectral structures; and I-Net, based on a UNet backbone,

which employs a scaling attention unit (SAU) to capture local and global implicit correlations while accelerating inference. To further enhance denoising, we introduce a multi-modal fusion unit (M-MFU) that effectively integrates multi-level features. Extensive experiments on various image types show that PGTNet surpasses both model-based and data-driven methods in subjective visual quality and objective quantitative measures.

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